

① 오답.md 어제 오답부터 재풀이 ② 신규는 공부계획 ID대로 ③ 채점: solve-exam-prep 로 풀이 대조

떠올릴 개념

- $\omega = 2\pi f$, 진폭·주기·위상
- $rms = peak / \sqrt{2}$ (정현파)
- $Z_R=R$, $Z_L=+j\omega L$, $Z_C= j/(\omega C)$

문항 목차 (총 2문항)

- [1] 9.1 — 정현파·phasor·임피던스
- [2] 9.3 — 정현파·phasor·임피던스

9.1 A sinusoidal current is given by the expression

$$i = 100 \cos (600t + 45^\circ) \text{ mA}.$$

Find (a) f in hertz; (b) T in milliseconds; (c) I_m ; (d) $i(0)$; (e) ϕ in degrees and radians; (f) the smallest positive value of t at which $i = 0$; and (g) the smallest positive value of t at which $di/dt = 0$.

— 풀이 —

9.3 Consider the sinusoidal voltage

$$v(t) = 25 \cos(400\pi t + 60^\circ) \text{ V}.$$

- a) What is the maximum amplitude of the voltage?
- b) What is the frequency in hertz?
- c) What is the frequency in radians per second?
- d) What is the phase angle in radians?
- e) What is the phase angle in degrees?
- f) What is the period in milliseconds?
- g) What is the first time after $t = 0$ that $v = 0$ V?
- h) The sinusoidal function is shifted $5/6$ ms to the right along the time axis. What is the expression for $v(t)$?
- i) What is the minimum number of milliseconds that the function must be shifted to the left if the expression for $v(t)$ is $25 \sin 400\pi t$ V?